ECOMMERCE INSIGHTS AND OPTIMIZATION

Data Analytics Capstone Project — Student Assignment

Project Theme: Transform ecommerce transaction, product, shipping, and time-based sales data into measurable business insights and actionable recommendations.

Project Area	Data Analytics / Business Intelligence
Primary Tools	SQL, Python, Excel, Google BigQuery
Visualization Options	Python, Excel, Power BI or Tableau
Difficulty	Intermediate to Advanced
Recommended Duration	2–3 weeks
Student Output	Analysis notebook + SQL script + dashboard/report + recommendations

Instructor Use: This document is designed as a reusable student project brief. Students should work independently, document assumptions, validate results, and explain the business meaning of every major finding.

1. Project Overview

Ecommerce companies generate large volumes of data from orders, products, customers, shipping, pricing, and time-based demand. The objective of this capstone is to simulate a real analytics assignment in which students move from raw data to business recommendations.

Students will work with ecommerce order-item and product datasets. The analysis should answer three core business questions: which product categories generate the strongest commercial performance, how shipping characteristics influence cost and profitability, and how sales vary over time.

The project is intentionally structured so that students practice the full analytics lifecycle: understanding the business problem, inspecting raw data, cleaning and validating data, joining datasets, writing SQL, performing exploratory analysis in Python, creating visualizations, identifying patterns, and communicating recommendations.

The final analysis must not stop at charts. Every major observation should be translated into a business implication and, where possible, a measurable action.

Business scenario

You are working as a Junior Data Analyst for an online retail company. Management wants to understand which products and categories deserve greater attention, which shipping characteristics are creating cost pressure, and when demand is strongest. The management team expects a concise but evidence-based recommendation rather than a collection of disconnected charts.

> Your task is to create an analytics solution that management could use for marketing, inventory planning, shipping optimization, and profitability improvement.

2. Learning Objectives

• Understand a business problem and translate it into measurable analytical questions.

• Inspect, clean, validate, and document ecommerce datasets.

• Design and use relationships between order-level and product-level data.

• Write SQL queries using filtering, aggregation, joins, CASE expressions, subqueries, CTEs, and window functions.

• Use Google BigQuery or another SQL environment to process analytical data.

• Use Python and pandas for exploratory data analysis and visualization.

• Calculate revenue, order metrics, average order value, shipping metrics, and category-level performance indicators.

• Analyze seasonal and monthly sales patterns.

• Investigate whether product weight, size, or category is associated with higher shipping costs.

• Convert analytical findings into actionable business recommendations.

• Present limitations, assumptions, data-quality issues, and opportunities for future predictive analytics.

3. Business Questions

Area	Primary Question	Example Metrics
Category Performance	Which categories and products contribute most to sales value?	Revenue, units, orders, AOV, revenue share
Shipping	Which products/categories create high shipping-cost exposure?	Shipping cost, cost per unit, shipping-to-sales ratio, weight
Seasonality	When does demand increase or decrease?	Monthly sales, monthly orders, growth rate, peak month

Area	Primary Question	Example Metrics
Optimization	What actions could improve profitability or operational efficiency?	Potential savings, promotion opportunities, inventory actions

4. Dataset and Project Files

The project is based on two primary CSV files. Students should inspect the actual files before making assumptions about field names, data types, missing values, or business definitions.

File	Purpose	Typical Analytical Use
order-items.csv	Order-line transaction data	Revenue, quantities, order dates, order/product relationships, shipping a
products.csv	Product master data	Product category, product attributes, price, weight, dimensions, shipping-
ERD Diagram.png	Database structure reference	Understanding relationships and keys
cis_122.sql	SQL analysis reference/starter	SQL-based extraction and analysis
Insights and Recommendations.pdf	Reference output	Expected style of business interpretation

Important student rule

Do not assume that a column named price is automatically revenue. Confirm whether price is a product list price, sale price, unit price, or another measure. Similarly, determine whether shipping cost is stored per order, per line item, or per product. Your metric definitions must be documented before analysis.

Suggested data dictionary template

Field	Table	Data Type	Business Meaning	Quality Check
order_id	order-items	Integer/Text	Unique identifier for an order	Check duplicates and nulls
product_id	Both	Integer/Text	Identifier linking order lines to product master	Check unmatched IDs
order_date	order-items	Date/DateTime	Date on which order was placed	Check valid dates
quantity	order-items	Numeric	Units purchased on a line	Check zero/negative values
price	Both/depending on file	Numeric	Price-related field	Confirm definition
shipping_cost	Order-items/depending on file	Numeric	Shipping charge/cost	Check negative/outliers
category	products	Text	Product category	Check blanks/standardization
weight	products	Numeric	Product weight	Check units and extreme values

5. Required Data Preparation

Before calculating any business metric, complete a documented data-preparation phase. The preparation should be reproducible rather than performed manually without notes.

1. Load both CSV files and inspect row counts, column names, data types, and sample records.

2. Identify primary keys and possible foreign-key relationships.

3. Check for missing values in critical fields such as product ID, order ID, date, price, quantity, category, and shipping fields.

4. Check for duplicate rows and duplicate business keys.

5. Check for negative, zero, or unusually large quantities and monetary values.

6. Standardize category names, date formats, and numeric data types.

7. Confirm units for weight and dimensions before interpreting them.

8. Join order-items with products using the appropriate product key.

9. Create a clean analytical table or view and document the transformation logic.

10. Validate the cleaned dataset by comparing totals before and after transformations.

Data-quality questions students must answer

• How many rows are present before cleaning and after cleaning?

• How many records contain missing values?

• How many product IDs in order data do not exist in the product table?

• Are there duplicate order-product combinations?

• Are there negative or zero sales values?

• Are shipping costs unusually high for a small number of records?

• Are all dates within a logical business period?

• Which assumptions were necessary because the dataset does not explicitly define a field?

6. SQL Analysis Requirements

SQL is the primary analytical language for structured aggregation. Students should create one clean SQL script containing clearly labeled sections. The script must be executable from top to bottom after the required tables are loaded.

Minimum SQL topics

• SELECT, WHERE, ORDER BY, GROUP BY, HAVING

• COUNT, SUM, AVG, MIN, MAX

• INNER JOIN and LEFT JOIN

• CASE WHEN for business classifications

• Common Table Expressions (CTEs)

• Subqueries

• Date functions for month/quarter/year analysis

• Window functions such as RANK, DENSE_RANK, LAG, or SUM OVER

• Safe division and null handling

• Conditional aggregation

Starter SQL exercises

The following examples are intentionally generic. Students must adapt column names after inspecting the actual dataset.

```
-- 1. Basic data inspection
SELECT *
FROM order_items
LIMIT 20;

-- 2. Revenue by category
SELECT
    p.category,
    SUM(oi.quantity * oi.price) AS revenue,
    SUM(oi.quantity) AS units
FROM order_items oi
JOIN products p
  ON oi.product_id = p.product_id
GROUP BY p.category
ORDER BY revenue DESC;

-- 3. Monthly sales
SELECT
    EXTRACT(YEAR FROM order_date) AS year,
    EXTRACT(MONTH FROM order_date) AS month,
    SUM(quantity * price) AS revenue
FROM order_items
GROUP BY year, month
ORDER BY year, month;

-- 4. Shipping analysis
SELECT
    p.category,
    AVG(oi.shipping_cost) AS avg_shipping_cost,
    SUM(oi.shipping_cost) AS total_shipping_cost
FROM order_items oi
JOIN products p
  ON oi.product_id = p.product_id
GROUP BY p.category
```

```
ORDER BY total_shipping_cost DESC;
```

Required SQL questions

1. Find the top 10 product categories by revenue.

2. Find the top 20 products by total sales value.

3. Calculate total units sold by category.

4. Calculate average selling price by category.

5. Calculate total and average shipping cost by category.

6. Identify products with high shipping cost relative to sales value.

7. Calculate monthly revenue and monthly order count.

8. Identify the highest-revenue month.

9. Calculate month-over-month revenue change using LAG.

10. Rank categories within each year using a window function.

11. Find products that contribute a disproportionately large share of revenue.

12. Create a final analytical view that combines product, sales, and shipping attributes.

Advanced SQL challenge

Create a category performance scorecard containing revenue, units sold, unique orders, average order value or average line value (depending on data granularity), total shipping cost, shipping-to-sales ratio, and revenue rank. Explain why each metric is useful and whether the metric should be interpreted as a product, order, or category measure.

7. Python Exploratory Data Analysis

Use Python, pandas, NumPy, matplotlib, and optionally seaborn to perform exploratory analysis. The notebook should be written as a story: question → code → output → interpretation → business implication.

Recommended notebook structure

Section	Expected Content
1. Setup	Imports, display options, file paths
2. Load Data	Read CSV files and verify successful loading
3. Data Audit	Shape, dtypes, nulls, duplicates, descriptive statistics
4. Cleaning	Missing values, type conversion, standardization, outliers
5. Merge	Join transaction and product information
6. KPI Creation	Revenue, units, orders, shipping, ratios
7. Category EDA	Category comparisons and concentration
8. Shipping EDA	Weight, shipping cost, cost ratios, outliers
9. Seasonal EDA	Month, quarter, year and trend analysis
10. Visualization	Business-focused charts with titles and labels
11. Findings	Evidence-based insights
12. Recommendations	Actions linked to findings

Recommended Python starter code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

orders = pd.read_csv("order-items.csv")
products = pd.read_csv("products.csv")

print("Orders shape:", orders.shape)
print("Products shape:", products.shape)

display(orders.head())
display(products.head())

print("\nMissing values:")
print(orders.isna().sum())

print("\nDuplicate rows:")
print("Orders:", orders.duplicated().sum())
print("Products:", products.duplicated().sum())

# After confirming column names:
merged = orders.merge(
    products,
    on="product_id",
    how="left",
    suffixes=("_order", "_product")
)

print("Merged shape:", merged.shape)
display(merged.head())
```

Required visualizations

- Bar chart: revenue by product category.
- Bar chart: units sold by category.
- Top-10 product revenue chart.
- Line chart: monthly revenue trend.
- Line chart: monthly order count.
- Scatter plot: product weight vs. shipping cost.
- Scatter plot: product price/revenue vs. shipping cost.
- Box plot or equivalent: shipping cost distribution by category.
- Heatmap or matrix: month versus category revenue, if sufficient data exists.
- Pareto-style chart showing cumulative revenue contribution of products or categories.

Visualization quality checklist

- Every chart has a descriptive title.
- Axes have meaningful labels and units.
- Currency values use consistent formatting.
- Categories are readable and not unnecessarily crowded.
- Charts answer a business question rather than simply displaying data.
- Outliers are investigated instead of silently removed.
- The notebook contains a written interpretation immediately after important charts.

8. Shipping Cost Analysis

Shipping analysis should go beyond identifying the category with the largest absolute shipping cost. A category can have a high total shipping cost simply because it sells many units. Students must therefore use normalized metrics as well.

Suggested shipping metrics

Metric	Formula / Definition	Interpretation
Total Shipping Cost	SUM(shipping cost)	Overall shipping exposure
Average Shipping Cost	AVG(shipping cost)	Typical shipping cost per analyzed record
Shipping Cost per Unit	Shipping cost / quantity	Normalizes shipping by volume
Shipping-to-Sales Ratio	Shipping cost / sales value	Shows shipping burden relative to sales
Weight	Product weight	Potential physical driver of shipping cost
Shipping Cost per Weight	Shipping cost / weight	Useful where weight is reliable and units are known

Student investigation

1. Identify the 10 products with the highest total shipping cost.

2. Identify the 10 products with the highest shipping-to-sales ratio.

3. Compare average shipping cost across categories.

4. Test whether heavier products generally have higher shipping costs.

5. Investigate whether high-priced products necessarily have high shipping costs.

6. Identify potential outliers and explain whether they appear to be data issues or legitimate business cases.

7. Propose at least three shipping optimization actions.

Important interpretation rule

Correlation does not prove causation. If a scatter plot shows that heavier products tend to have higher shipping costs, describe this as an observed association unless you have a valid causal design. Consider additional variables such as dimensions, destination, carrier, service level, or product category if those fields exist.

9. Seasonal Sales Analysis

Seasonality analysis examines how demand changes across time. Students should identify recurring patterns where the available date range is sufficient. If only a short period is available, clearly state that seasonal conclusions are limited.

• Create year, quarter, month, and month-name fields.

• Calculate monthly revenue and order volume.

• Identify peak and low-demand periods.

• Compare category behavior across months.

• Calculate month-over-month change.

• If multiple years exist, compare the same month across years.

• Separate genuine seasonality from one-time spikes caused by unusual events or data issues.

- Translate demand patterns into inventory and marketing planning recommendations.

10. Excel Analysis Component

Excel may be used for initial exploration, validation, pivot analysis, or a management-friendly summary. Students should demonstrate that they can reproduce at least a subset of the analysis using spreadsheet techniques.

Minimum Excel tasks

• Import the cleaned dataset or analytical output.

• Create a PivotTable for category revenue.

• Create a PivotTable for monthly revenue.

• Create a PivotTable for shipping cost by category.

• Use formulas such as SUMIFS, COUNTIFS, AVERAGEIFS, XLOOKUP or INDEX/MATCH where appropriate.

• Create a small management dashboard with 4–6 KPIs and 3–5 charts.

• Use data validation or structured tables where useful.

• Document the source and refresh process.

11. Dashboard Requirements

Students may build the final dashboard in Excel, Power BI, or Tableau. The dashboard should be designed for a manager who has only a few minutes to understand the situation.

Recommended KPI cards

KPI	Purpose
Total Revenue	Overall sales value in the analyzed period
Total Units	Volume sold
Total Orders	Transaction/order activity
Average Order Value	Average value per order where order-level calculation is supported
Total Shipping Cost	Shipping expenditure/exposure
Shipping-to-Sales Ratio	Shipping burden relative to sales
Top Category	Category with highest defined sales metric
Peak Sales Month	Month with highest revenue

Dashboard interaction ideas

• Date/year filter.

• Category filter.

• Top-N product selector.

• Monthly trend view.

• Shipping analysis view.

• Drill-down from category to product.

• Clear KPI definitions displayed on the dashboard.

12. Business Insight Framework

Each insight should follow a structured format so that the analysis remains actionable.

Step	Example Structure
Observation	Category A generated the highest revenue in the analyzed period.
Evidence	Revenue = ■X; revenue share = Y%; units = Z.
Interpretation	The category is a major contributor to sales and may warrant continued commercial attention.
Action	Evaluate promotion, assortment, pricing, or inventory strategy.
Measurement	Track revenue, margin, conversion, inventory availability, and repeat performance.

13. Recommendations Framework

Students must provide recommendations that are directly connected to evidence. Avoid generic statements such as 'increase marketing' without explaining what should be marketed, when, why, and how success will be measured.

Recommendation areas

• **Product strategy:** Focus attention on high-contribution categories/products while investigating weak performers.

• **Marketing:** Align campaigns with categories and periods that demonstrate measurable demand.

• **Inventory:** Use observed monthly demand patterns to improve stock planning.

• **Shipping:** Investigate packaging, carrier, service level, minimum-order thresholds, or fulfillment strategies for high-cost products.

• **Pricing:** Examine whether shipping burden or product price affects the commercial attractiveness of selected products.

• **Data quality:** Establish validation rules for future data ingestion.

Recommendation template

Problem → Evidence → Root-cause hypothesis → Recommended action → Owner → KPI → Expected impact → Measurement period.

14. Advanced Analytics Extensions

Students who complete the core project can extend it into a predictive analytics project.

• Monthly sales forecasting using moving averages, exponential smoothing, Prophet, ARIMA, or another justified method.

• Product demand forecasting.

• Category-level demand forecasting.

• Customer segmentation if customer-level data becomes available.

• Product clustering using price, weight, category, sales volume, and shipping characteristics.

• Anomaly detection for unusual sales or shipping records.

• Regression analysis to study relationships between product characteristics and shipping cost.

• Scenario analysis estimating the effect of shipping-cost reductions.

• Recommendation systems for product cross-sell or upsell opportunities if customer/order history is available.

Predictive-modeling rule

Do not build a model merely to include machine learning. A model must answer a clearly defined business question, have a target variable, use an appropriate evaluation metric, and explain how the result would be used by the business.

15. BigQuery Implementation

Students using Google BigQuery should create a dataset/project structure that is easy to reproduce. Suggested structure: raw tables → cleaned tables/views → analytical tables → reporting queries.

• Load CSV files into separate raw tables.

- Confirm schema and data types.
- Create cleaned views or tables.
- Use SQL transformations rather than manual edits where possible.
- Create reusable analytical views for category, shipping, and seasonal analysis.
- Use partitioning/clustering where appropriate for larger datasets.
- Document table names, relationships, and transformation logic.
- Capture screenshots or export results for the final report.

16. Student Deliverables

Deliverable	Requirement
1. SQL Script	Clean, commented SQL covering required analytical questions
2. Python Notebook	Data loading, cleaning, EDA, visualizations, interpretations
3. Excel File	Pivot analysis and management summary
4. Dashboard	Excel, Power BI, or Tableau dashboard
5. Final Report	Business findings, evidence, recommendations, limitations
6. Data Dictionary	Definitions, data types, assumptions, quality issues
7. Presentation	5–10 minute explanation of business problem, findings, and actions

17. Suggested Assessment Rubric — 100 Marks

Component	Marks	Assessment Focus
Data Understanding & Cleaning	15	Data profiling, quality checks, documented assumptions
SQL Analysis	20	Correct joins, aggregation, CTEs, window functions, business logic
Python EDA	15	pandas workflow, calculations, reproducibility, interpretation
Visualization / Dashboard	15	Clarity, relevance, usability, correct chart selection
Business Insights	15	Evidence-based findings and meaningful interpretation
Recommendations	10	Actionability, measurement, connection to evidence
Documentation & Presentation	10	Structure, communication, professionalism

18. Student Evaluation Questions

1. What are the three most important findings from your analysis?

2. Which category has the strongest sales performance, and what metric did you use?

3. How concentrated is revenue among the top products or categories?

4. Which products or categories have the highest shipping burden?

5. Does product weight appear associated with shipping cost?

6. Which month or period has the highest demand?

7. What data-quality issue had the greatest potential impact on your results?

8. What assumptions did you make about revenue, orders, shipping, or product attributes?

9. Which recommendation should management implement first based on your evidence, and how would you measure its impact?

10. What additional data would improve your analysis?

11. If you had three more months, what predictive or optimization model would you build?

12. How would you explain your most important chart to a non-technical manager?

19. Common Mistakes to Avoid

- Joining tables without validating the join key.
- Double-counting revenue because of duplicated product or order records.
- Calling a line-item average an average order value.
- Using total shipping cost without normalizing for sales or units.
- Removing outliers without investigating why they exist.
- Treating correlation as proof of causation.
- Creating charts without a business question.
- Reporting numbers without explaining what they mean.
- Making recommendations that are not supported by the data.
- Changing raw data directly without retaining an original copy.

20. Recommended Project Timeline

Phase	Suggested Time	Activities
Phase 1	Day 1–2	Business understanding, file inspection, data dictionary
Phase 2	Day 3–4	Cleaning, validation, ERD/key verification
Phase 3	Day 5–7	SQL analysis and KPI creation
Phase 4	Day 8–10	Python EDA and visualizations
Phase 5	Day 11–12	Excel/dashboard development
Phase 6	Day 13–14	Insights, recommendations, final report
Phase 7	Day 15	Presentation, peer review, corrections

21. Final Presentation Structure

1. Business problem and objective.

2. Dataset overview and data-quality summary.

3. Analytical approach and tools.

4. Top category/product findings.

5. Shipping-cost findings.

6. Seasonal sales findings.

7. Dashboard walkthrough.

8. Three to five most important recommendations.

9. Limitations and assumptions.

10. Future analytics roadmap.

22. Suggested Viva / Interview Questions

• Why did you choose SQL for the first stage of analysis?

• Why is a LEFT JOIN useful during data-quality validation?

• What happens if one product ID appears multiple times in the product table?

• What is the difference between revenue, profit, and margin?

• Why might total shipping cost be misleading when comparing categories?

• When should you use a window function instead of GROUP BY?

• What is the difference between a trend and seasonality?

• How would you handle missing product weights?

• How would you validate your monthly revenue calculation?

• How would you explain an outlier to a business stakeholder?

• What additional fields would you request from the logistics team?

• How would you convert your analysis into an automated reporting pipeline?

23. Final Submission Checklist

- ■ Raw files retained and not overwritten.
- ■ Data dictionary completed.
- ■ Cleaning steps documented.
- ■ Join keys validated.
- ■ SQL script executes without unexplained errors.
- ■ Python notebook executes from top to bottom.
- ■ Required charts are included and interpreted.
- ■ Dashboard contains clearly defined KPIs.
- ■ At least five evidence-based insights are documented.
- ■ At least five actionable recommendations are documented.
- ■ Limitations and assumptions are explicitly stated.
- ■ Final report is proofread.
- ■ Presentation is ready.
- ■ All files use clear names and are organized in folders.

24. Suggested Folder Structure

```
Ecommerce_Analytics_Project/
■
■■■ data/
■   ■■■ raw/
■   ■   ■■■ order-items.csv
■   ■   ■■■ products.csv
■   ■■■ cleaned/
■
■■■ sql/
■   ■■■ ecommerce_analysis.sql
■
■■■ python/
■   ■■■ ecommerce_eda.ipynb
■
■■■ excel/
■   ■■■ ecommerce_analysis.xlsx
■
■■■ dashboard/
■   ■■■ ecommerce_dashboard.pbix   # or Tableau/Excel output
■
■■■ report/
■   ■■■ final_report.pdf
■
■■■ presentation/
■   ■■■ final_presentation.pptx
■
■■■ README.md
```

25. Capstone Challenge — Management Optimization Scenario

Assume management gives you the following instruction: *"We need to improve ecommerce profitability without reducing customer demand. Use the available data to identify where we should focus."*

Your final solution should build a defensible analytical argument around this instruction. You are not expected to calculate a perfect profit model if profit or margin data is unavailable. Instead, identify measurable drivers that can reasonably be connected to profitability, such as sales contribution, shipping burden, demand concentration, and seasonal inventory risk.

Required challenge output

1. Build a category-level performance table.

2. Build a product-level performance table.

3. Identify high-revenue and high-shipping-burden segments.

4. Quantify the importance of the top categories/products using revenue share.

5. Identify peak and low-demand periods.

6. Create at least one analysis connecting product characteristics to shipping cost.

7. Provide a prioritized action plan with measurable KPIs.

8. Explain what additional data is required to calculate true profitability rather than sales-based performance.

26. Extension: Profitability Thinking

If the dataset does not contain cost of goods sold, gross margin, discounts, returns, taxes, payment fees, or fulfillment costs, students should explicitly state that revenue is not the same as profit. A category with high revenue may have low or negative profitability if its costs are high.

Students may therefore create a **profitability proxy** rather than claiming to calculate actual profit. For example, a proxy can combine revenue contribution and shipping burden, but it must be clearly labeled as a proxy and not as accounting profit.

27. Expected Final Insight Style

A strong final report should read like a business analysis rather than a programming assignment. For example:

Finding: A small number of categories account for a large share of observed revenue.
Evidence: The top categories contribute X% of revenue in the analyzed period.
Implication: Changes in availability or promotion within these categories could materially affect overall sales.
Action: Review inventory coverage and promotional planning for these categories.
KPI: Track category revenue share, stock-out rate, conversion, and campaign ROI.

Students should replace placeholders such as X% with their calculated values and provide the relevant chart or table as evidence.

28. Instructor Notes

• Encourage students to explain calculations verbally rather than reading code.

• Ask students to defend metric definitions, especially revenue and order-level metrics.

• Require students to demonstrate at least one SQL query live.

• Ask students to reproduce one result in Python or Excel as a validation exercise.

• Review joins carefully because incorrect joins can silently produce inflated totals.

• Use the advanced challenges to differentiate stronger students.

• For team projects, require individual ownership of a defined analytical component.

• Ask students to document limitations rather than penalizing them for missing data that genuinely does not exist.

29. Project Completion Standard

A project is considered complete when the student can move from raw ecommerce data to a reproducible analytical output, explain the calculations, defend the findings with evidence, communicate the business implications, and propose measurable actions. The final submission should demonstrate both technical data-analytics capability and business communication.

End of Student Project Brief